Identifying Hallucinations in Sequences Generated from the ESM2 Protein Language Model

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Abstract

In recent years, as protein language models (PLMs) have grown in complexity, function, and ubiquity, a persistent problem has been the question of interpretability and quality assurance of the sequences and structures they predict. Many *in silico* predictions of proteins have lacked biological activity when utilised in a laboratory environment. A key driver of this, are hallucinations, in which models confidently predict sequences and structures that are predicted to have some kind of advantageous feature or optimisation within a model, but possess implausible sequences and subsequent folds. This study identified a series of features across a set of sequences that were purposefully hallucinated by overrunning a directed evolution model with ESM-2 to convergence, then identifying traits of these models. It was found that the hallucinated proteins broadly had increased thermostability whilst having reduced contacts, itself a contradictory finding, and homorepeat motifs were frequently found throughout sequences, whilst motifs common in thermophiles were rarely found. The subsequent structures predicted from sequences had poor coverage across Foldseek, confirming that a key hallmark of hallucinations are localised regions of implausible folds. The findings of this study set a foundation for the identification of drivers of hallucinations and study into the decision making of PLMs via study of their knowledge neurons and using this to inform classification models for evaluating structures for implausible folds.

1. Introduction

The advent of machine learning has revolutionised protein engineering, allowing for rapid iteration and proposals for proteins that serve a variety of needs, such as medicine, agriculture, or industry, by altering proteins to exhibit or improve upon desirable functional properties, or by creating entirely new enzymes. A variety of approaches have been utilised to this end, such as directed evolution, semi-rational or rational design, or *de novo* design (Bose, 2024). These approaches have recently been augmented by the advent of protein language models (PLMs), which have allowed for computational prediction of more promising candidates or routes for protein engineering, or even outright novel proteins with little resemblance to training data via the use of diffusion models (Watson et al 2023). However, an issue within the usage of these methods for protein engineering is that models are capable of hallucinating implausible sequences and structures. Whilst hallucinations can be desirable, particularly in diffusion models for the purposes of *de novo* protein design, in many situations, predictions are made that result in a structure with no actual biological activity. This paper seeks to identify features of hallucinations in PLMs, and subsequent structures inferred from sequence. In this study, it is prudent to define how the term 'hallucination' is used – the term can indeed be used in a positive framing, particularly in the context of diffusion models. But for clarity, in this paper, the term 'hallucination' refers to a PLM confidently predicting a sequence or structure which lacks biological activity, analogous to hallucinations seen in language models used for text generation resulting in incorrect output that is nonetheless predicted as a likely response to a query by the model. In addition, this paper also focuses on hallucinations within generated sequences – however, the quality in structural predictions is also observed, as in many sequence to structure models, structure is inferred as a function of sequence.

This project set out to initially generate a set of hallucinated protein sequences to identify features that a model optimises for, then foster an understanding of what a model considers a "perfect" sequence, when the output is controlled to be an implausible structure. This is followed by analysis of the quality of structures predicted using the model. ESM2 was utilised as a sequence prediction model, using a directed evolution tool to iteratively mutate a seed sequence to purposefully induce hallucinations, and examine how the predicted physiochemical properties of the variant differed from the wild-type, identifying features the model optimises for when it is hallucinating. The purpose of this analysis is to identify characteristics of hallucinations in sequence and structural prediction models, to lay the groundwork for further research in drivers of hallucinations, and from the results and proof of concept laid out in this paper, lay the foundations for building classification tools that can predict whether a given sequence structure is hallucinated.

2. Literature Review

PLMs have advanced exponentially in terms of scale and capability in the past few years. These models have largely been dominated by the utilisation of the transformer architecture introduced by Vaswani et al in 2017. Meta's ESM series leveraged evolutionary data to infer structure from sequence, due to the use of sequences that occurred in nature producing realistic sequences and folds. ESM-1b (Rives et al 2021) was released in 2020 with 650 million parameters, and capable of learning features that define structure and biological activity from a language model. Building on from this, ESM2, was released ranging from 8 million to 15 billion parameters could predict 3-dimensional structure with its accompanying ESMFold, built on the embeddings generated in ESM2 (Lin et al 2023). ESM-3 has been developed with 98 billion parameters, with the impressive capability of predicting structure from multimodal prompts such as information about solvent-accessible surface area or structural co-ordinates and even generating a novel green-fluorescent protein from these prompts. (Hayes et al, 2024). A key development to note however, is that as PLMs have advanced, they have gained the ability to infer structure from sequence, largely in part due to this information being contained within evolutionary history, and structural information and motifs being encoded within protein sequences as a result. What this theoretically enables, is the prediction of novel sequences which then should in turn, result in realistic sequences – this however, remains an imperfect science. This remains a pertinent issue that is gripping the field: many predictions that result in sequences and structures that lack biological activity *in vitro.* This is demonstrated by Repecka et al (2019), who found that only 24% of generated sequences from a protein sequence prediction tool were soluble and possessed catalytic activity. Whilst even 24% is still a remarkable achievement that has the potential to result in significantly higher throughput of candidate proteins in research and industrial settings, it still represents a lack of accuracy that remains a problem for sequence and structure prediction models. A significant driver for inaccuracies structural models is that they have been trained on experimentally determined structural data – the structures yielded by methods such as X-ray crystallography do not necessarily represent the native state and behaviour of real a protein in solution. This does not account for all the inaccuracy in structural prediction however. Both sequence and structure prediction models can generate hallucinations and artefacts in their output, most notably folds that are not seen in nature.

Further compounding the issue of implausible sequences and structures, is the lack of interpretability in many models. It is not fully understood how transformer models make the decisions they do. In the context of PLMs, efforts have been made in this area: Schreiber (2023) identified how amino acid substitutions at a certain point in a sequence altered the log-likelihood of a protein. Hou et al (2025) demonstrated that ESM2 performed better with the mid-sized, 650 million parameter model rather than the 15 billion on ProteinGYM, itself a benchmarking tool for protein fitness prediction (Notin et al 2023). An implication of Hou et al's research was that larger parameter models had a collapse of dynamic range of log-likelihood ratios, often predicting negative scores for mutations proteins that have been experimentally determined to have a neutral effect on biological activity. It was suggested that as a model approaches a 'perfect' perplexity of 1, it fails to differentiate between deleterious and neutral mutations and having reduced specificity for functionally important residues. Schreiber's approach to analysing mutational effect, and Hao et al's choice of ESM2 650M model has heavily informed the methodology within this paper.

The lack of interpretability in PLMs however leads to ethical implications, particularly if transformers are used in decision making in situations with an extremely low tolerance for failure (Sato & Takagi, 2025). By identifying drivers of decision making in transformers, it would allow for greater accountability and interpretability, in addition to being able to identify when outputs are hallucinated, misleading, or erroneous. In the context of protein structure and sequence prediction, this is particularly pertinent. Indeed, whilst human language or images can be easier to parse for error by observation or applying domain knowledge to quickly identify erroneous outputs, the same process is more difficult for protein structures and sequences predicted by machine-learning methods – aside from glaringly obvious anomalies, we cannot ascertain that a predicted sequence or structure will yield a viable protein in reality. Furthermore, the high throughput environment of predictions these are used in within the laboratory often makes manual validation unviable, necessitating methods of evaluation for veracity that are scalable and usable in high throughput.

Improving the reliability, quality of output, and interpretability of PLMs has been an active area of research. In this literature review, an overview of two major areas of research are laid out: evaluation metrics, where efforts are made to scrutinise and quality assure the predictions of models, and mechanisms of hallucinations: understanding why models are confidently predicting implausible results.

Evaluation Metrics

As research into PLMs for this use expands, as does the need for evaluation metrics that can scrutinise robustly the outputs of predicted sequences and structures. Broadly, research into evaluation metrics uses an approach of analysing the physiochemical or structural qualities of sequences and/or structures, or the identification of features associated with poor quality. Physiochemical evaluation metrics are of protein sequence and structure are well-established and remain in use today. Grand average of hydropathicity (GRAVY) was introduced by Kyle & Dolittle (1983), taking an average of the hydropathy, the affinity for water, for every amino acid in a sequence. Ramachandran outliers (Ramachandran et al, 1963) have been used as a means of determining whether bond angles in the protein backbone are realistic. Alignment-based metrics using substitution matrices such as BLOSUM62 (Henikoff & Henikoff, 1992) utilise sequence identity to real sequences, using their similarity to sequences that exist within evolutionary. Machine learning methods may be trained on assay labelled data, in which function has been observed and annotated from laboratory settings, or even combinations of assay-labelled and evolutionary data (Hsu et al 2022). Alphafold2 utilises pLDDT, a measure of local confidence within a structure, providing a measure of a the congruence that the predicted model would have with an experimental structure (Jumper et al 2022). However, pLDDT has a limitation in that it scores disordered and flexible regions with a lower score, despite these being perfectly natural and normal occurrences in a functional protein. Johnson et al (2024) developed a computational scoring pipeline that improved the rate of experimental success in over 500 enzymes that were expressed from generative predictions by 50-150%. The authors note that it is necessary to account for a wide variety of factors when designing computational metrics, due to the multitude of factors that can account for the poor activity of a predicted protein in biological settings, and this is reflected in the pipeline they built, composite metrics for protein sequence selection (COMPSS). This study found that inverse folding models and language model LLRs were the most effective means of predicting biological activity in predicted protein structures of the methods used. Foldseek (van Kempen et al 2024) has proven to be an effective means of evaluating structures, by representing structures as strings of 3-dimensional interactions, known as 3Di strings. By utilising TM-score, a measure of how well a query has aligned to a target, in conjunction with E-values, good scores in these metrics against folds found in nature can provide confidence that a predicted fold is a realistic and viable structure. By abstracting structural information into strings, Foldseek enables sensitive searches of structural information many orders of magnitude than previous, equally sensitive methods. What is common with all these methods though, is they all have their own strengths and shortcomings. As a result, it is often necessary to corroborate a series of metrics together when assessing the viability of sequences and structures.

Mechanisms of Hallucination

Ji et al (2023) describes hallucinations within LLMs as producing outputs that are not faithful to the corpus of the input data when recovering and generating information from it. There remains a gap in the research in specifically identifying hallucinations in PLMs, however. Efforts have been made to evaluate output of PLMs, but not specifically identifying features and drivers of hallucinations, which is a key research question this paper seeks to answer.

A notable mechanism of hallucination identified by Ji et al is known as degeneration, when output is bland, incoherent, or gets stuck in loops. While this description was written in relation to textual language models, this mechanism certainly appears in proteins. Jin et al categorised hallucinations as intrinsic hallucinations: hallucinations that contradict source content, and extrinsic hallucinations: hallucinations that cannot be verified from source content. Both of which occur in the output of PLMs. Farquhar et al (2024) proposed a method of identifying hallucinations within LLMs: by using semantic entropy. This involves clustering possible answers to a question by similar meaning, and looking at the likelihoods of all the possible answers, determining that an answer with low entropy is one the LLM is more confident in. Whilst this method is more difficult with PLMs, strategies such as nearest neighbour approaches and creating a proposal distribution of next amino acid substitutions or identifying knowledge neurons for key information can drastically reduce search space and potentially allow such methods to be adapted for scrutinising the decision making of PLMs. Lindsey et al (2025) developed a means of abstracting a simplified decision-making process of a language model by creating a more human readable, simplified model of it, allowing for the creation of attribution graphs approximating the original model. In this study, it was proposed that hallucinations are rooted in models being incentivised for making plausible guesses for queries on poorly represented information in the model, in conjunction with "misfires" in a circuit which gives a "can't answer" answer in response to a query. It is also possible for models to misattribute due to similarity in concepts a token represents, an example of "Dallas" and "Austin" being conflated in a query about the capital of Texas. Before applying this context to the language of PLMs however, it is necessary to understand the representations in models. A rather reductive analogy for the conflation of Texan capitals would be the information flow in a model query being: "create a salt bridge between position i and j". The correct answer would be a positively and negatively charged amino acid in a pairing. However, hallucinations could be possible in the form of assigning both positions with the same charge residue in the case of a conflation, or an outright wrong one, depending on how the model has learned the representation. Whilst this may not be necessarily how PLMs make decisions, it illustrates with a human readable analogy for something that has extremely complex and abstract decision making, that is often learned on enormous datasets of evolutionary data. What these approaches inform though, are the steps to be taken after this study, and the goals for what this study seeks to achieve: reducing the search space and identifying hallmarks of hallucination, so adapting methods for identifying hallucinations becomes more viable.

3. Method

Method overview

To identify drivers of hallucination, it was necessary to generate sequences, and their respective structures in which there was a high level of confidence that they were hallucinations - these outputs would not exist as viable proteins. The methodology laid out here approaches this task by utilising a directed evolution method that repetitively attempts to improve a model's output of likelihood of a sequence in respect to the seed sequence, and running this until the model converges and cannot improve the likelihood of the sequence anymore. The outputs of this process then have structures predicted, with both predicted sequences and structure then measured for a variety of evaluation metrics and physiochemical properties. From this, it can be interrogated as to what features models are optimising for, to inform knowledge neuron attribution and semantic entropy studies.

Protein Family Selection

A series of proteins from three families, kinases, dehydrogenases, and proteases were selected. These families were selected due to being well characterised, with a large number of high-quality structures available. Families were selected on Interprot, with reviewed, experimentally determined structures only. Proteins were further filtered by resolution, selecting those only with a resolution equal to or less than 2.0 Ångstroms.

Selecting Most Distant Sequences

To identify the most distant protein sequences within each family, fasta files were embedded into an embedding space for each separate family using ESM2's *esm2_t33_650M_UR50D* tokeniser. After embeddings were generated, K-means was used as a clustering method where the 10 most maximally distant centroids were identified, and the sequence closest to the centroids were selected for downstream use. The results were visualised using t-SNE as a dimensionality reduction technique using a consistent random seed to ensure reproducibility, This results in 30 initial seed sequences overall, ten from each family. An example of the visualisation of this clustering is shown on the dehydrogenase family below (Figure 3.1).

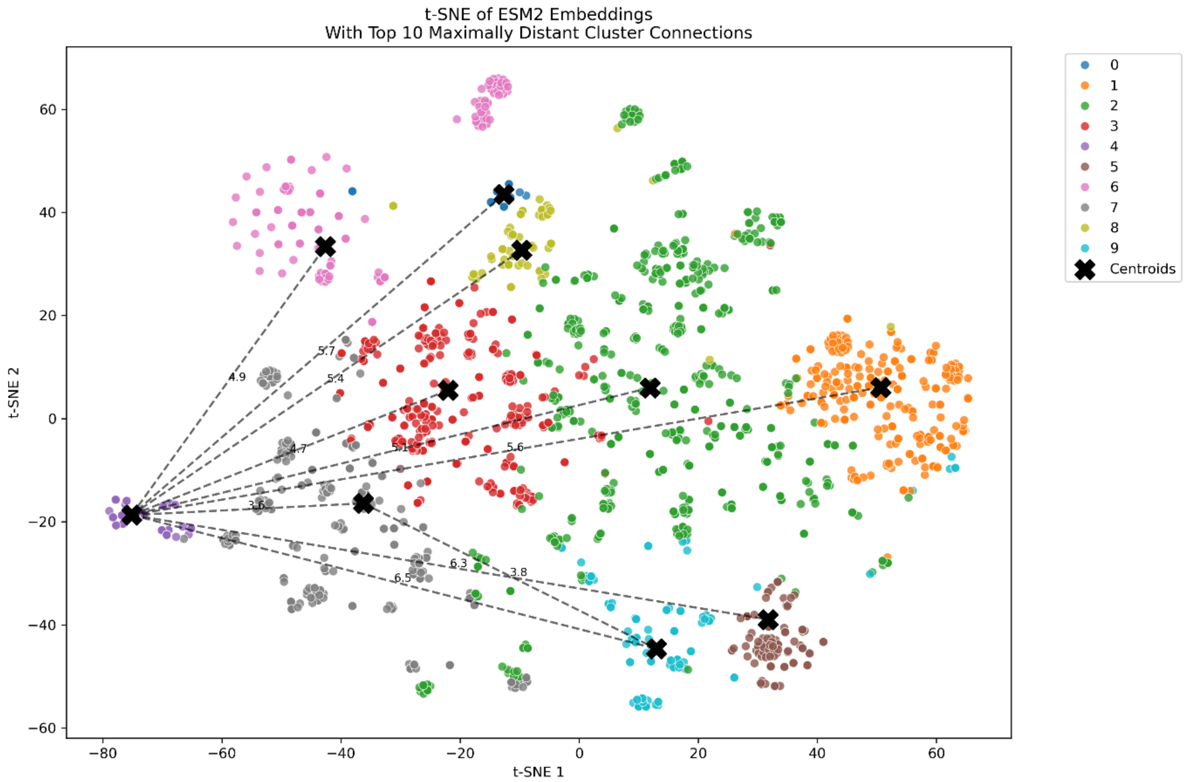


Figure 3.1 t-SNE visualisation of protein family clusters for dehydrogenases projected onto a two-dimensional space.

Directed Evolution and Hallucination Generation

To purposefully generate hallucinations, an adaptation of EvoProtGrad was used (Emami et al 2023). EvoProtGrad is a gradient-based discrete Markov Chain Monte Carlo (MCMC) method for directed evolution. EvoProtGrad functions by utilising a 'product of experts' (PoE) score, (the product being the product of log-likelihood of sequences from each 'expert' model that contributes to prediction) that is sampled from a proposal distribution of the highest scoring sequences for log-likelihood for a sequence if n amino acids were changed (Figure 3.2). The PoE is given as the equation:

$$P(x) = F(x)G(x)$$

Where *F(x)* is the proposal distribution of the output of unsupervised models, and *G(x)* is the proposal distribution of the output of supervised models. The probability for the entire sequence can be calculated by multiplying the probability of every token in the sequence, allowing for relative probabilities to be assigned lists of candidate protein sequences.

Figure 3.2. Output of EvoProtGrad in console. Note the PoE score changing in respect to the wild-type sequence in chain 0, as mutations are proposed (letters in red).

EvoProtGrad avoids unnecessary computation by only proposing mutations from a local neighbourhood, that is, only the likelihood of sequences that can be approached by mutating n amino acids. By using only one model as an expert for EvoProtGrad, it is possible to isolate what a single model is continually optimising for, which whilst is not in line with the intended use of EvoProtGrad to improve the fitness of inputted proteins, was appropriate for this study.

The *esm2_t33_650M_UR50D* model was used as the expert. Scoring method was set as 'pseudo log-likelihood ratio'. For each sequence, an initialisation step was performed, in which a maximum of eight mutations were proposed at each MCMC step, with four parallel Markov chains, for 100 MCMC steps. After this, the directed evolution instance was run for 50 MCMC steps with four parallel chains, and a maximum of four proposed mutations. The sequence from each iteration with the highest PoE score was then extracted and used as the seed sequence for the next iteration of the directed evolution loop. This process was repeated until the model could no longer increase the PoE for the sequence, with the change in PoE converging (defined as best PoE from the last 10 directed evolution instance calls having an average score of 0.1 or less), after which, the next protein was selected, and the process repeated. A small number of sequences had errors and could not be repeated due to the timescale of the project, however.

Prediction of Structures of Hallucinated Sequences

To predict structures of hallucinated sequences, Alphafold2 was chosen for its ease of use and accessibility. Whilst ESMFold was a preferred option due to its speed and the use of ESM2 in the previous step, ESMFold cannot process sequences more than 400 amino acids long on the server-based tool. Furthermore, as Alphafold2 uses MSA to generate structures, it is hypothesised that hallucinated sequences in are likely to result in poor quality structures due to a lack of depth from similar sequences to align against.

For every structure, a four Alphafold2 structures were predicted. A structure of the final converged sequence, and structures for the 1^{st}, 5^{th} ,and 10^{th} sequences proposed by EvoProtGrad. These final three structures served as controls to compare quality degradation of structures as the EvoProtGrad process iterates, controlling for whether metrics of lower protein quality are due to the model being purposefully run to produce poor quality sequences, or simply due to being computationally generated, hence why sequences from the start of the run were used to create structures too.

Evaluation Metrics for Hallucinated Sequences

To identify features the model optimised for, sequences in this process, both wildtype and variant had a variety of physiochemical properties observed to identify significant differences between wild types and variants. Thermostability (melt temperature), instability index, and change in amino acid propensity. Thermostability was measured using TemBERTure (Rodella et al 2024), with the average of three replicates used as the melt temperature of the sequence. TemBERTure was run on the wild-type and variant, with a Taq polymerase as a thermostable control, and a K12 polymerase as a mesophilic control. Every intermediary MCMC proposed sequence was also measured. This allowed for tracking the change in thermostability as the model ran. Protparam was used within Biopython to measure the other properties. Within Protparam, the following methodologies are used; hydropathy is calculated using GRAVY, and instability index is a calculation used by Guruparad et al (1990).

Evaluation Metrics for Hallucinated Structures

The quality of hallucinated structures was measured by Molprobity, from Phenix (Chen et al, 2010). Clashscore was used to identify steric clashes (defined as number of clashes per 1,000 residues), for rapidly determining a measure of model quality that could be used to confirm whether repeated optimisation reduced model quality. pLDDT and SASA were not used as they were noted by Johnson et al that they were poorly predictive of enzyme activity. Contacts were counted, with the change in contacts within a model measured between wild-type and variant. Contacts were defined by heavy atom protein pairs with a 4.5Å cut-off, using the method proposed by Yuan et al (2012). Salt bridges were counted using Barlow & Thornton's (1983) definition: an ion pair 4.0Å or less apart.

Using Foldseek to Identify Hallucinations

Foldseek was used to identify areas of poor alignment in variants, comparing them to the PDB structures of the wild-types, and observing how well queries aligned to hits in the Alphafold-Swissprot database. The rationale behind this was to determine whether folds being generated were found in nature. The difference in the raw number of hits between wild-type and variant structures is also recorded, as this provides a measure of how well Foldseek was able to find hits for hallucinated structures as opposed to variants. Both local and global alignments were used, with K-mer filtering disabled, as to minimise the filtration of poor-quality hits, and retain areas of and poor quality for the sake of comparison. Foldseek easy-search was used, and the following commands were used:

Global TMalign

```
 --format-output
"query,target,taxid,taxname,evalue,bits,alnlen,qstart,qend,qlen,qcov,tcov,
alntmscore,qseq,tseq" \
  --alignment-type 2 \
  --prefilter-mode 1 \
```

Local: 3Di+AA Gotoh-Smith-Waterman

```
--format-output
"query,target,taxid,taxname,evalue,bits,alnlen,qstart,qend,qlen,qcov,tcov,
alntmscore,qseq,tseq" \
  --alignment-type 1 \
  --prefilter-mode 1 \
```

4. Results

Within these results, paired sample two tail t-tests were used to determine whether changes between wild-type and variants after directed evolution were statistically significant.

Changes in thermostability

The primary finding in regard to what ESM2 optimises for was that the predicted melt temperature of a protein almost always increased as a result of the process of directed evolution guided by the outputs of ESM2 (Figure 3.) ($p=5.39 \times 10^{-5}$). 2bhp and 2ejw are the only wild-type proteins that were thermophilic, and these had very small changes in melt temperature.

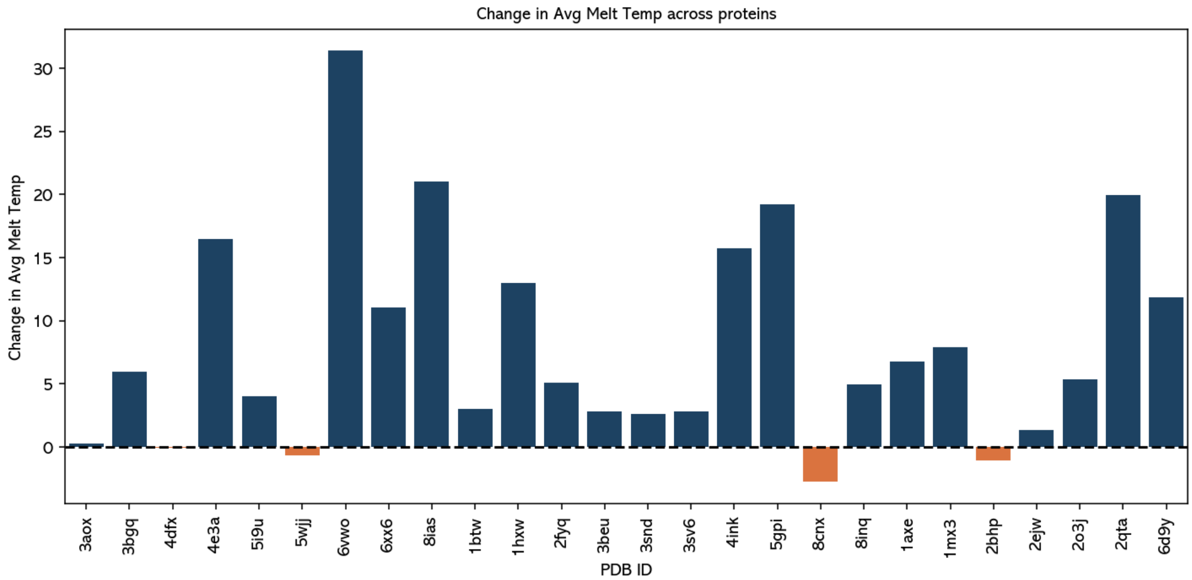


Figure 4.1 Change in average melt temperature across three TemBERTure replicates for each protein mutated.

In many cases, a peak of thermostability was met in early epochs. Within this selection of proteins, gradual linear increases, and late epoch increases in thermostability also occur, but there is not one defining pattern, other than the absolute melt temperature changing between the wild-type and final variant. The below graphs (Figure 4.2, Figure 4.3, Figure 4.4) plot the increase in thermostability and cumulative increase in log-likelihood as the model runs through epochs.

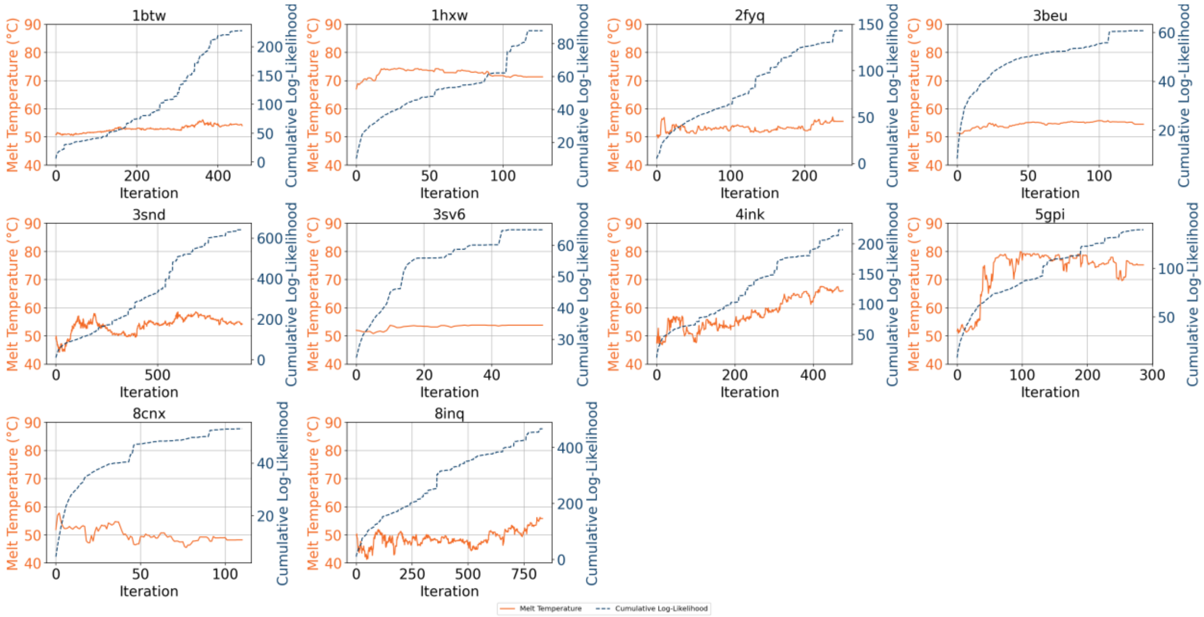


Figure 4.2 Protease thermostability vs cumulative log-likelihood.

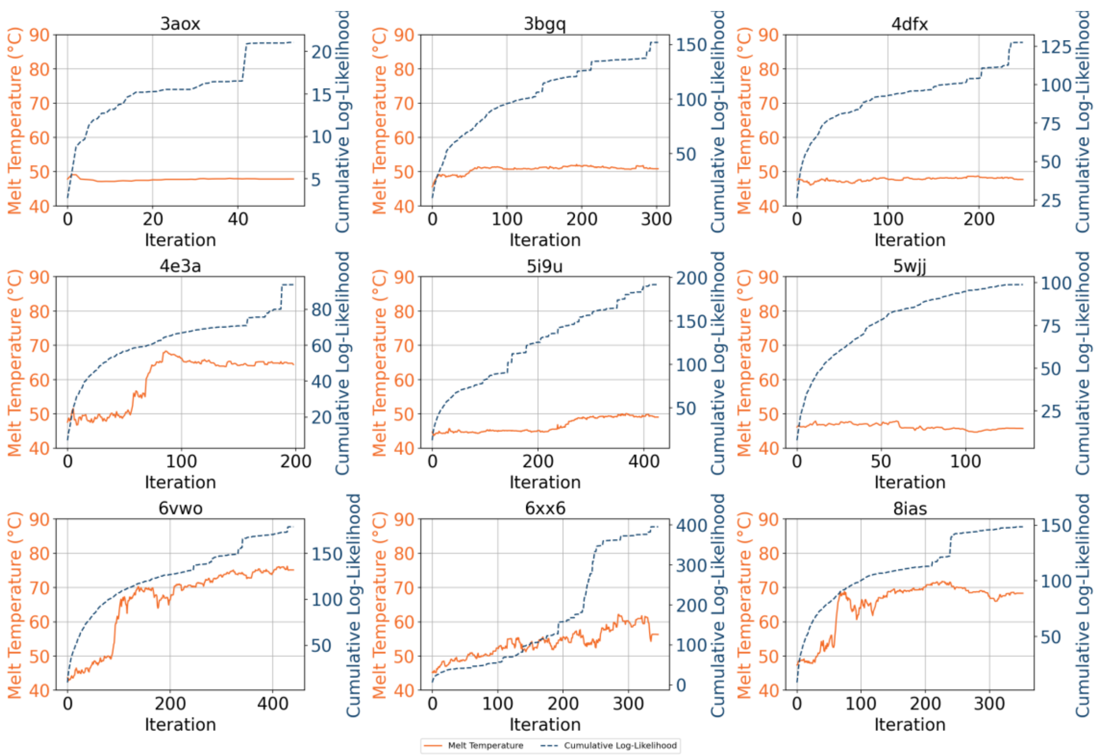


Figure 4.3 Kinase thermostability vs cumulative log-likelihood.

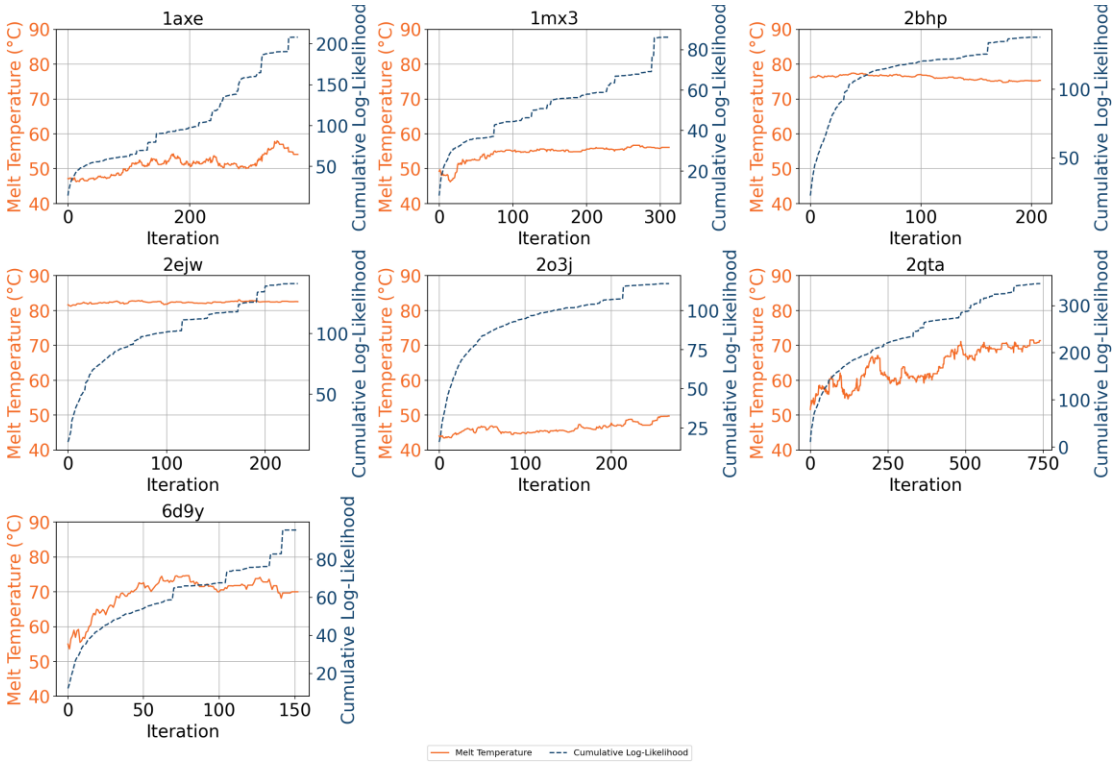


Figure 4.4 Dehydrogenase thermostability vs cumulative log-likelihood.

Amino acid counts and proportion

Changes across amino acid counts were observed. The below heatmap (Figure 4.5) demonstrates the most frequent substitutions are for lysine from arginine, leucine from isoleucine, valine from isoleucine, valine from leucine, aspartic acid to glutamic acid, and isoleucine from valine. Alanine is also a frequent target for substitutions across all amino acids.

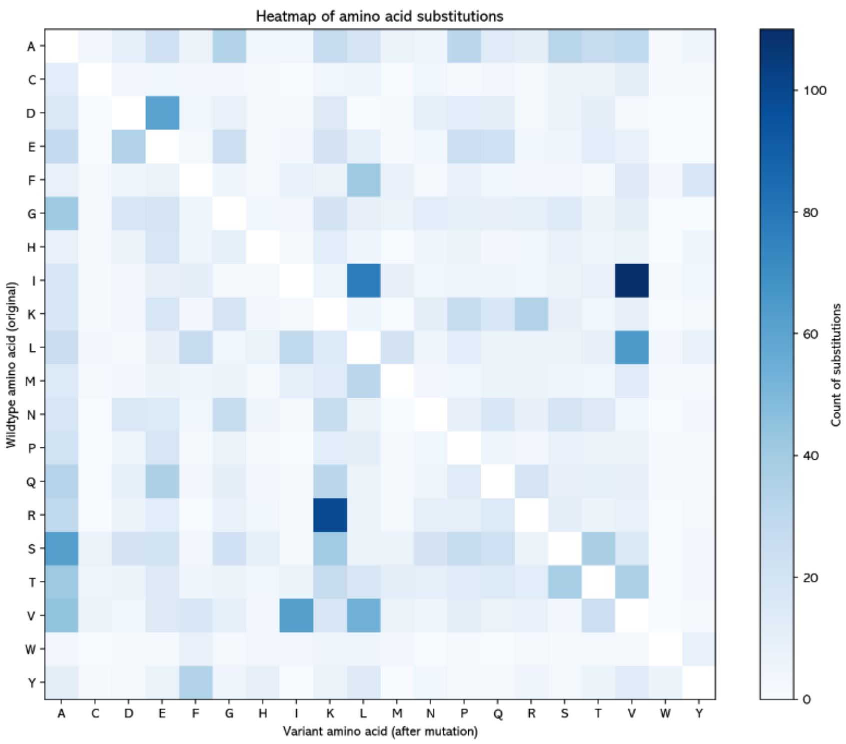


Figure 4.5. Counts of amino acid substitutions across entire set of proteins.

The below graph (Figure 4.6) demonstrates the absolute percentage change in amino acid proportions across sets. Whilst certain substitutions were commonly made between variant and wild-types, the end result is that changes in favour of alanine, lysine, glutamic acid, leucine, proline, and valine dominate the dataset of proteins used.

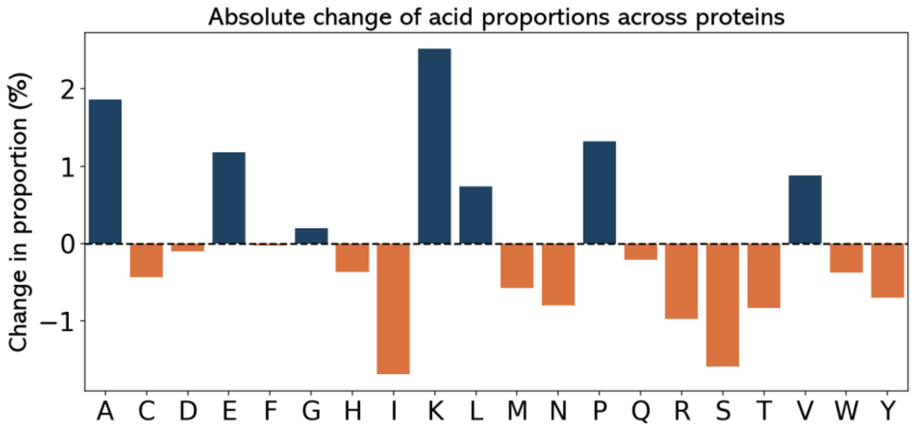


Figure 4.6. Absolute percentage change in amino acid substitutions across all samples.

Changes in Structure Quality

A decrease in the structure quality was observed as the model converged. Clashscore increased (R^2 = 0.313), suggesting that some element of the higher clash score from larger models is due to repeated iterations of predicted sequences producing lower quality structures (Figure 4.7). It should be caveated, that this may be skewed to an outlier effect, and would require further testing on a larger dataset, with the directed evolution loop set to not finish until high epochs (>600) have been reached, to generate a larger population size. At lower max epochs, the link between max epoch and clashscore is poorly correlated, suggesting perhaps that max epoch only becomes a more reliable predictor of model quality at extreme values.

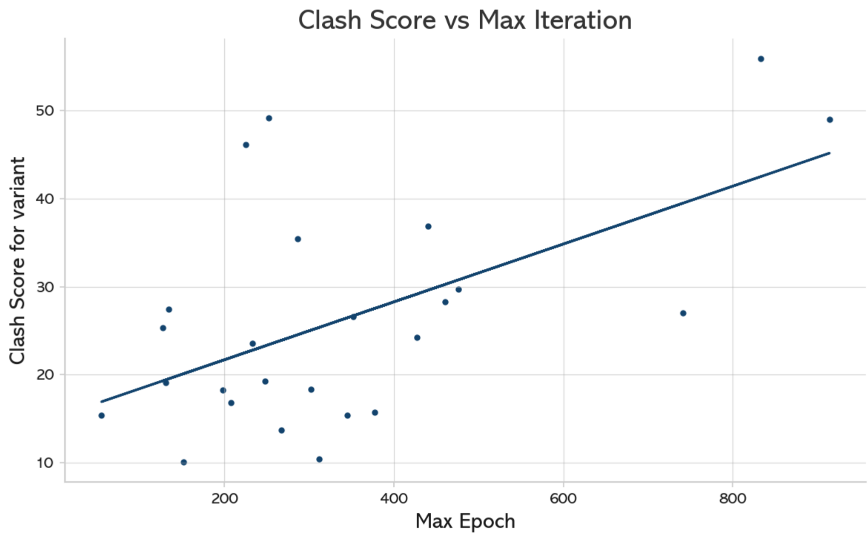


Figure 4.7. Linear regression of clash score against the final epoch number of every protein.

Running the model until convergence resulted in higher clash scores for predicted sequences

To measure whether running a model until convergence produces lower quality models, wild-type structures (wt), and Alphafold2 predicted models of early-stage epochs (1, 5, and 10) were used as controls to measure the variant against. All Alphafold2 predicted models of the variant had higher clash score than the wild type. There is a distinct difference in clash score between variants (labelled as var) and other predicted models. The increase in outliers on the box plot (Figure 4.8) harmonise with the observation that sequences with high final epochs result in higher clash scores. Pairwise comparison suggests that there is a significant difference between the group's variant and the wild type, 1 epoch and 5 epoch groups: wild type/variant: $p = 0.0$, 1/variant: $p = 0.0037$, 5/variant: $p = 0.0039$, 10/variant: $p = 0.1929$ (Figure 15).

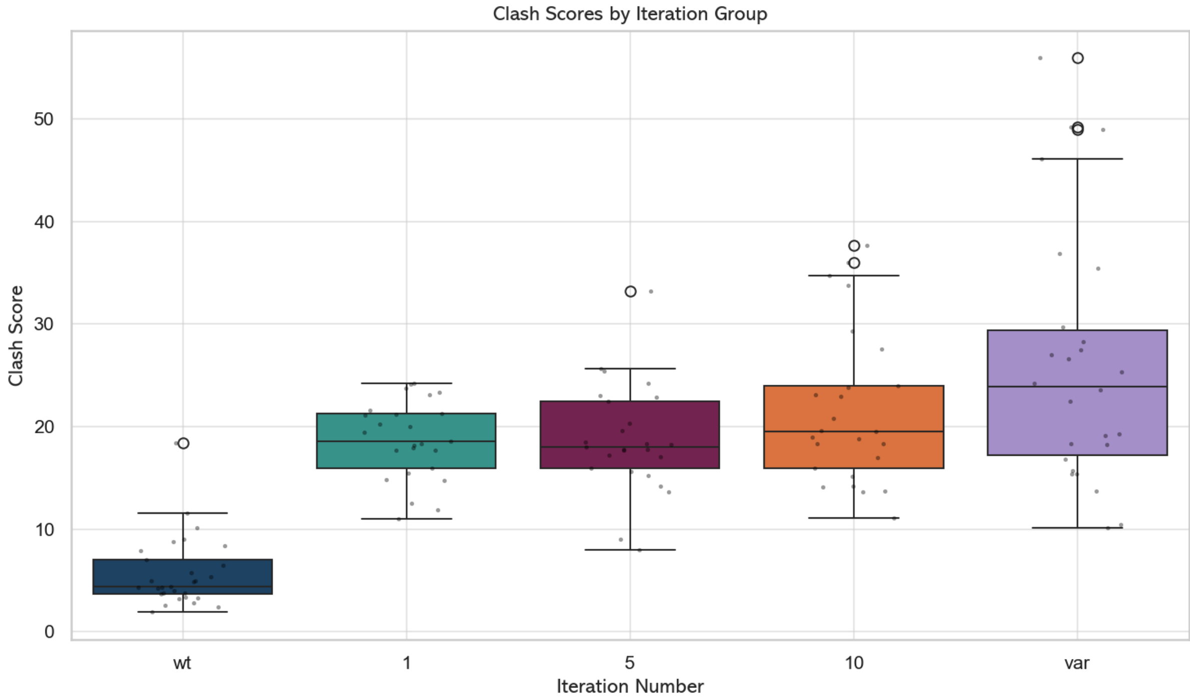


Figure 4.8. Box plot displaying distribution of clash scores of protein sequences for set epochs and the wild type and final variant sequences.

Mutated residues had lower contacts

On average, for each residue, mutations resulted in a reduced number of contacts ($p = 2.551 \times 10^{-31}$).

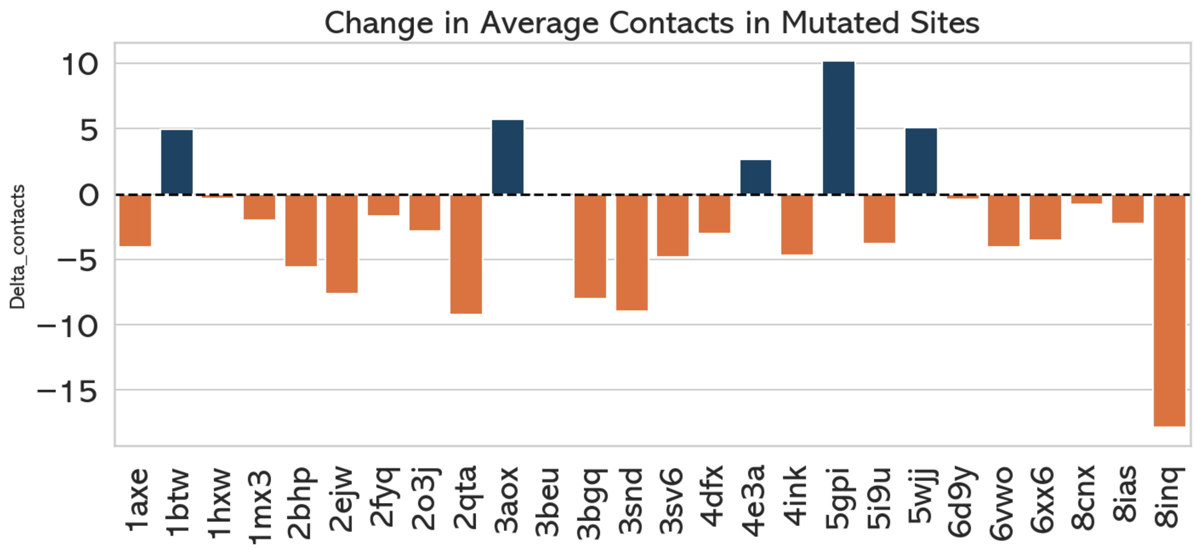


Figure 4.9. Changes in contacts on mutated residues, normalised for protein length.

Changes in salt bridge formation

Changes in salt bridge formation as observed (increase, n = 14/26; no change, n = 3/26, decrease, 9/26). All decreases in salt bridges occurred in proteins which also had decreased contacts.

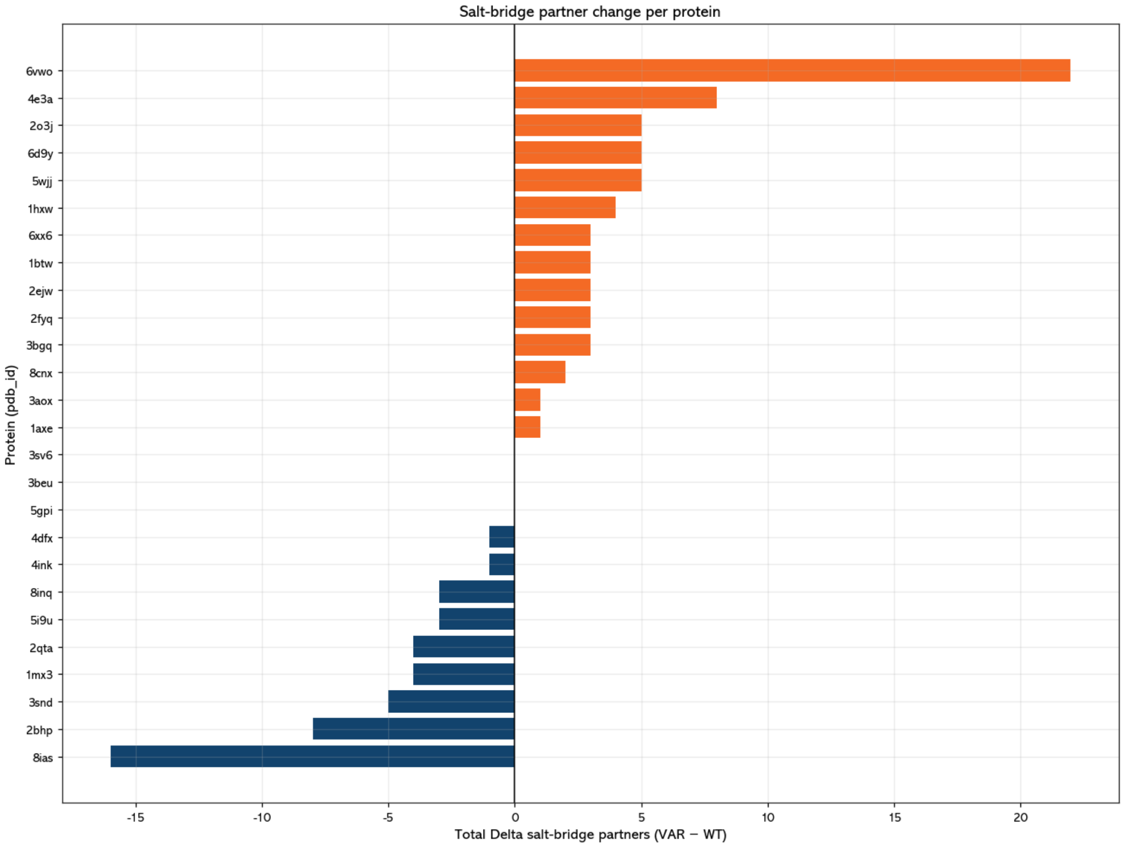


Figure 4.10. Changes in number of salt bridges per protein.

Only a few sequences had mutations focused on certain regions.

The below heatmap (Figure 4.11) demonstrates the position of mutations on a sequence by percentage through the length of the sequence. Aside from 3SV6, 5GPI, and 3AOX, there is no dominant position for mutations to consistently occur across sequences.

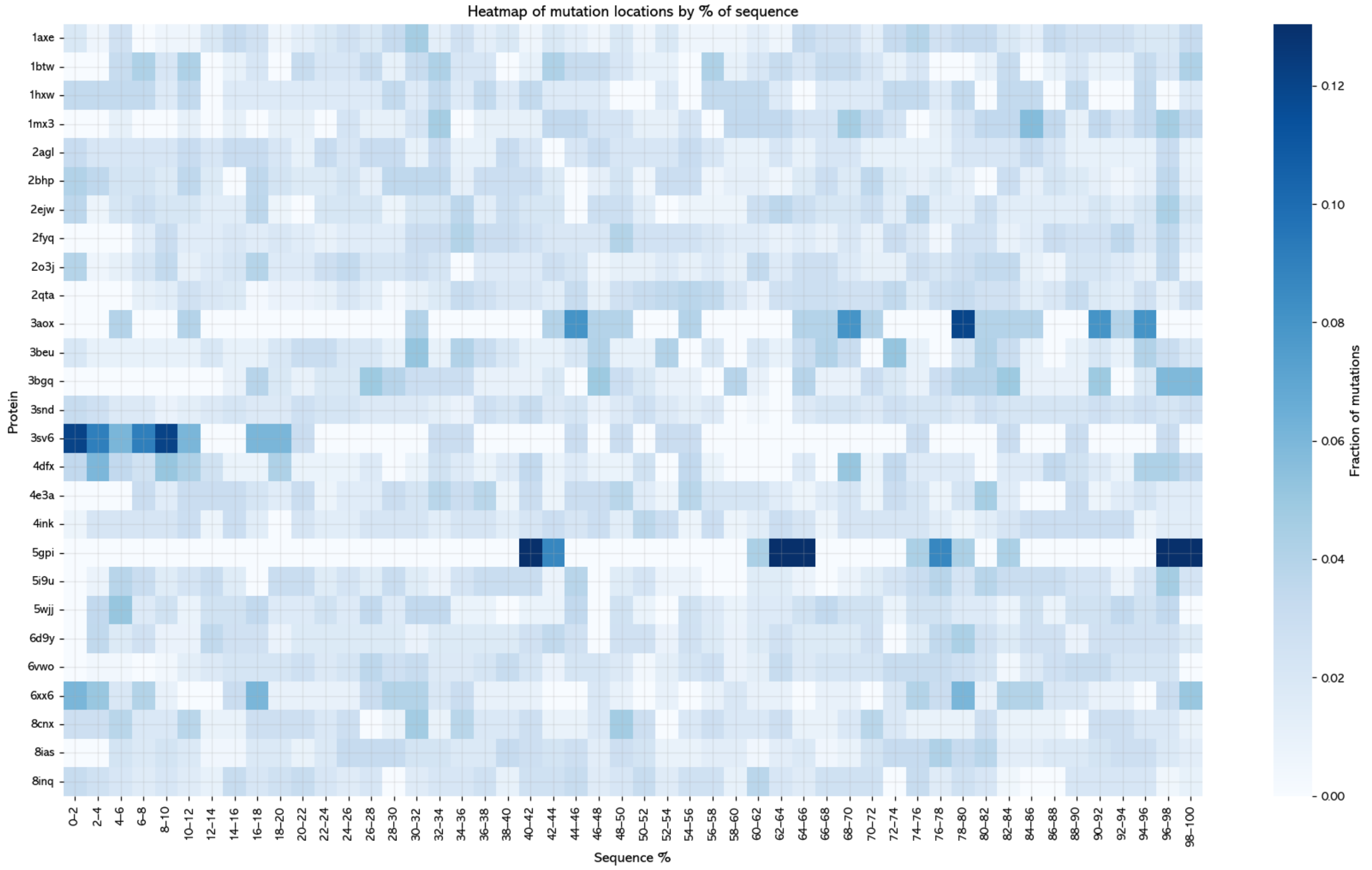


Figure 4.11. Location of where mutations occurred as a percent of total mutations in sequence.

Proposals to linkers were dominated by same residue motifs

A common set of linker proteins derived from Chen et al (2012), with triplet motifs additionally added from the amino acids observed to have increased in proportion across the dataset, P, A, K, E, V, G, L (AAAA was used instead of AAA to differentiate it from EAAAK, a common linker) were interrogated for their frequency. From this heatmap, it is evident that in many cases, the model optimises for residues by creating motifs of the same peptide. In addition, a series of common linkers were observed. Notably however, the common glycine-serine combination in linkers was reduced in almost all proteins (Figure 4.12).

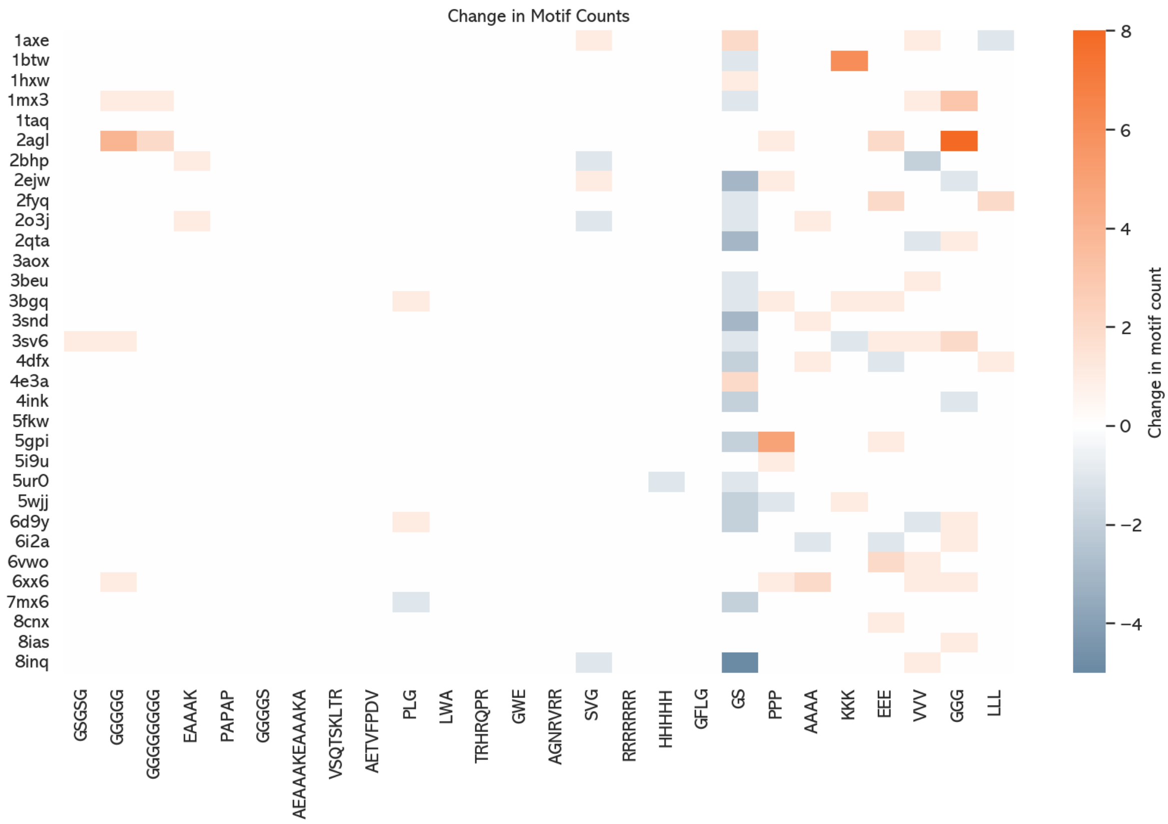


Figure 4.12. Change in counts of motifs of common linker proteins. Homorepeats dominate.

Amino acid propensities of thermostable proteins

Taylor and Vaisman (2010) identified the amino acid propensities of thermophilic, hyperthermophilic, and mesophilic proteins. The below chart outlines these, with the propensities of the wild-type and variant propensities from this study.

Propensity	A	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y
Mesophilic	8.26	1.92	5.8	6.53	3.85	7.29	2.29	5.48	6.22	8.86	2.17	4.51	4.52	4.04	4.79	6.07	5.62	6.9	1.43	3.44
Thermophilic	10.05	0.78	5.13	8.32	3.62	8.3	2.15	4.8	4.77	10.26	1.87	3.19	5.43	2.69	6.82	4.07	4.74	8.42	1.31	3.3
Hyperthermophilic	7.26	0.78	5.2	10.05	4.07	7.17	1.64	7.73	8.5	9.1	2.11	3.49	4.02	1.8	5.56	4.51	4	8.44	1.02	3.56
Wild-type	8.27	1.63	5.17	6.55	4.04	8.49	2.72	5.71	5.62	8.60	2.48	4.06	4.51	3.51	5.03	6.02	5.46	7.62	1.22	3.31
Variant	10.06	1.34	5.17	7.69	3.79	9.08	2.37	4.03	7.84	9.27	1.84	3.22	5.85	3.36	4.17	4.32	4.80	8.39	0.86	2.56

Figure 4.13. Amino acid propensities for varying sets of proteins.

The distance (L1 and L2) between each set, along with the count of close propensity pairs (defined as greater or less than 0.5%) is laid out in the graphs below (Figure 4.14).

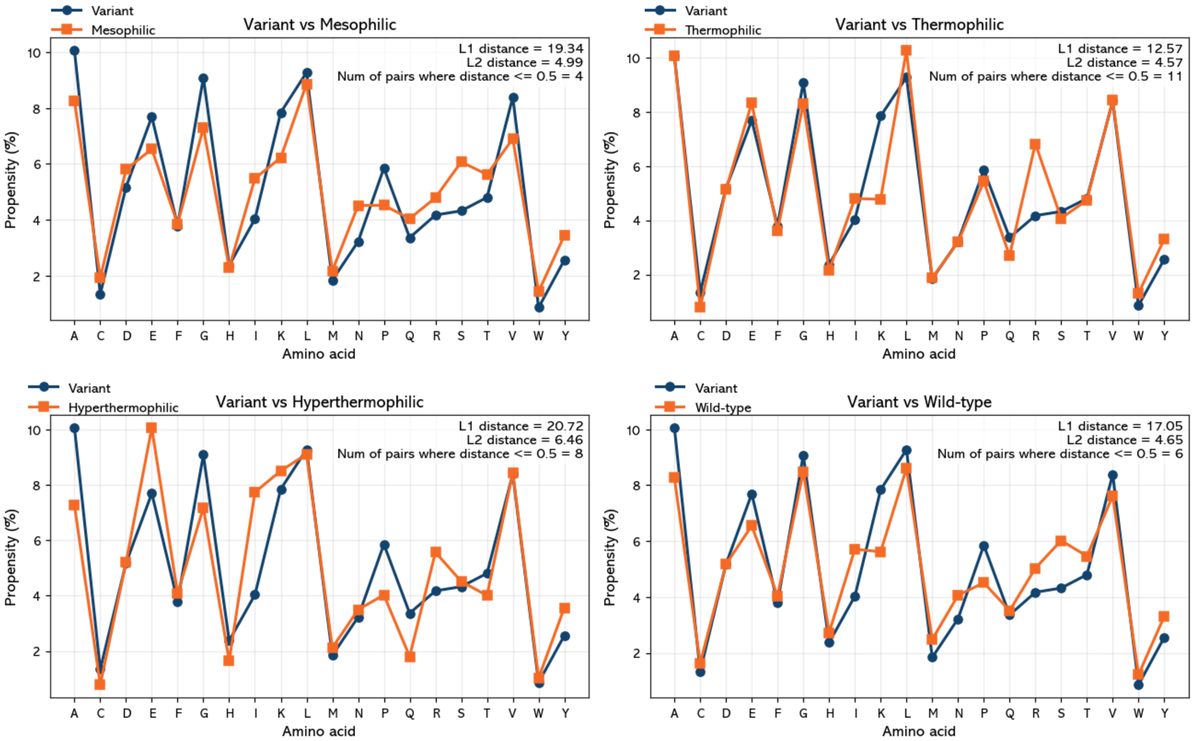


Figure 4.14. Amino acid propensity comparison of variant vs other sets. Similarity is greatest to thermophiles.

The amino acid propensities of the variant are closest to thermophilic amino propensities (distance = 20.83). There are highly similar propensities in some instances (Alanine: 10.05 in thermophilic, and 10.06 in variant), there are other instances in which the difference is vast, such as the case in lysine, which the variant misses significantly (7.84 in variant, 4.77 for thermophilic, 8.5 in hyperthermophilic). In addition to identifying amino acid propensities, Taylor and Vaisman also identified a series of 4-residue motifs that are overrepresented in thermophilic and hyperthermophilic proteins. The below graph (Figure 4.15) demonstrates the change in motif counts associated with residue quadruplets that are overrepresented in thermophiles and hyperthermophiles.

Figure 4.15. Changes the model had towards overrepresented motifs in thermophiles. Only EEEE had a significant change across proteins.

Notably, there does not appear to be common movement towards particular motifs. EEEE appears to be a common choice for the model to optimise towards – a highly represented amino acid in thermophiles.

The below chart (Figure 4.16) demonstrates a move towards homorepeat motifs. The change in the number of non-overlapping homorepeat motifs of n = 3-7 within each mutated sequence is plotted. M, V, A, K, P, E, L, R were chosen as they represent the amino acids that increased in proportion across the set. Histidine was also included due to the proclivity for the model to alter poly-H tails at the termini of sequences.

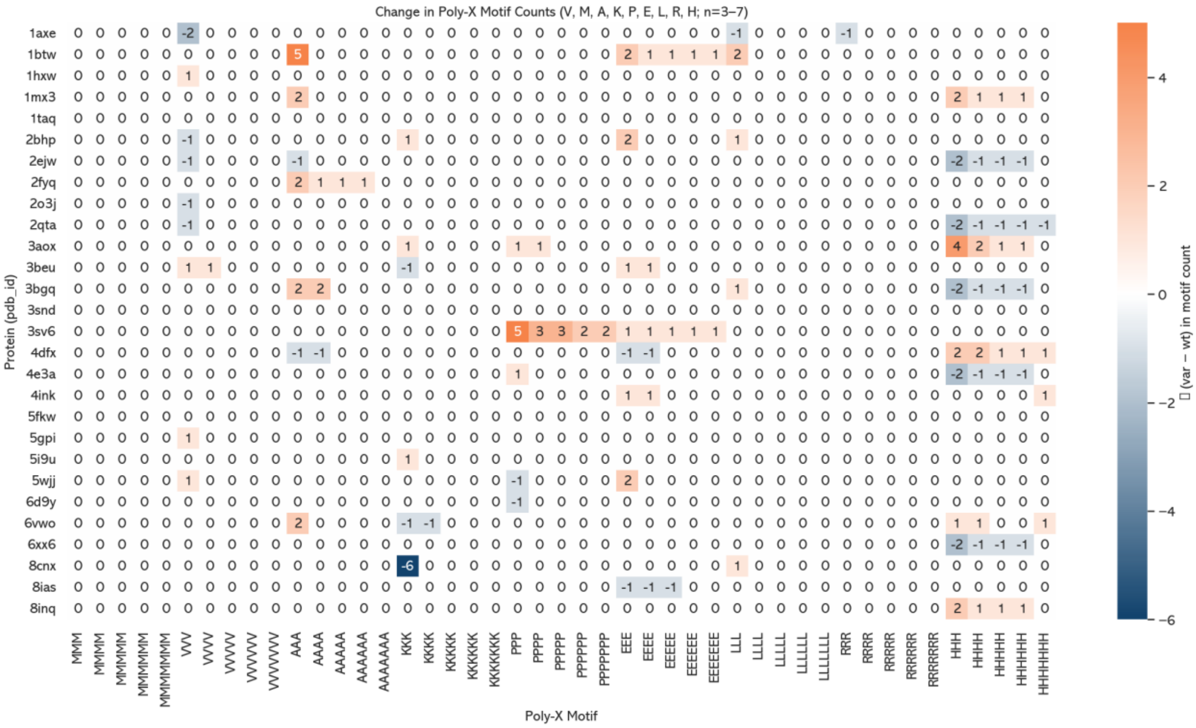


Figure 4.16. The change in homorepeats of increasing length.

Foldseek Global Alignment

The below chart (Figure 4.17) demonstrates the change in Foldseek hits for each query sequence, normalised by length of sequence. The overall trend is one of decline (positive change, n=3/26; no change, n=7/26, negative change, n = 16/26). Positive changes are of negligible magnitude.

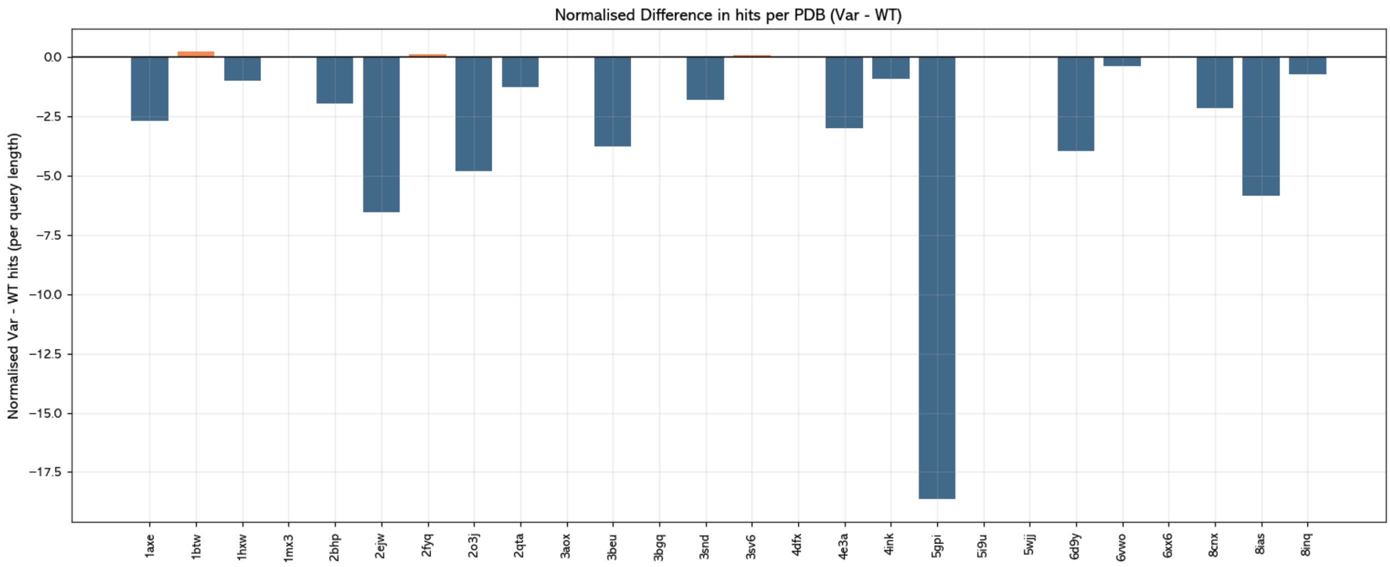


Figure 4.17. Change in hits, normalised as change in hits per residue.

The below charts (4.18) demonstrate the TM-alignment score of Foldseek across sequences normalised for length. Sequences are binned into 1% segments, and the average alignment score for all hits that cover that bin. Green lines represent the presence of homorepeats of at least 3 amino acids long. The third chart demonstrates the change in mean TM-score against each segment of the sequence.

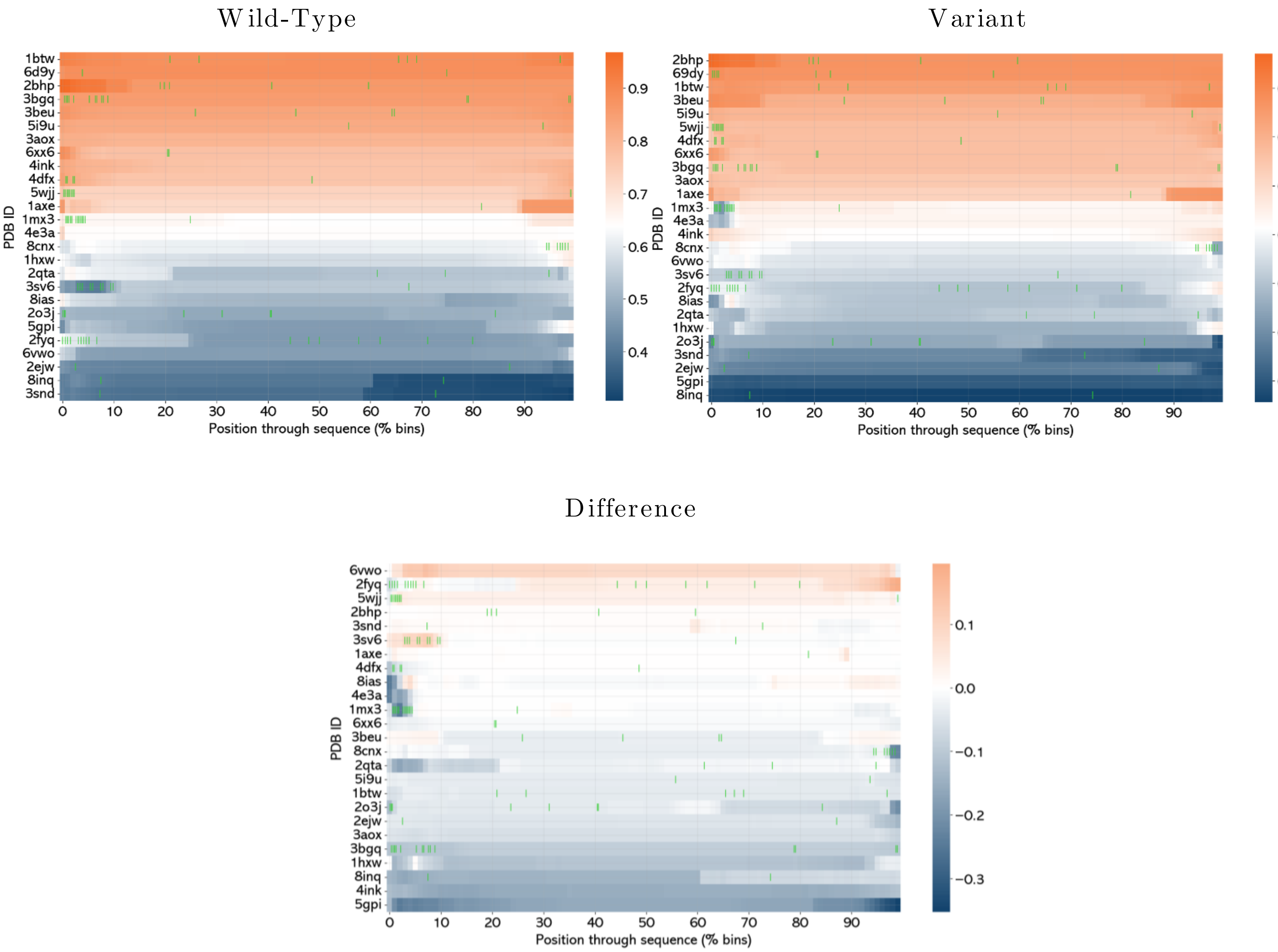


Figure 4.18. wild-type, variant, and difference heatmap of change in alignment quality (TM-score).

Foldseek Local Alignment

In local alignments (Figure 4.19), the overall trend in hits between wild-type and variant is one of decline (positive change, n=2/26; no change, n=7/26, negative change, n = 17/26)

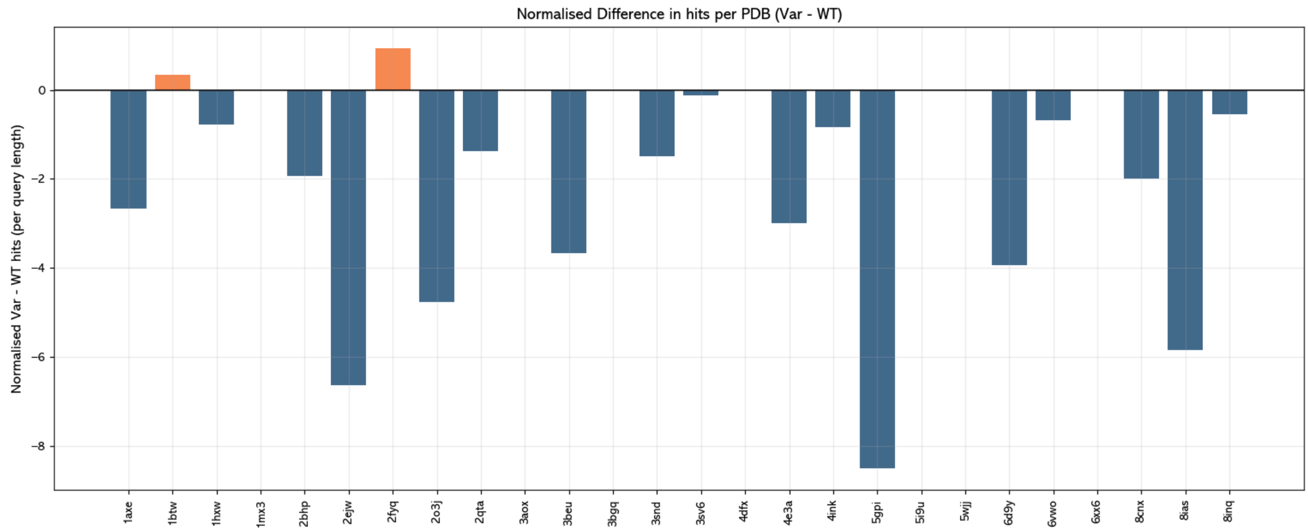
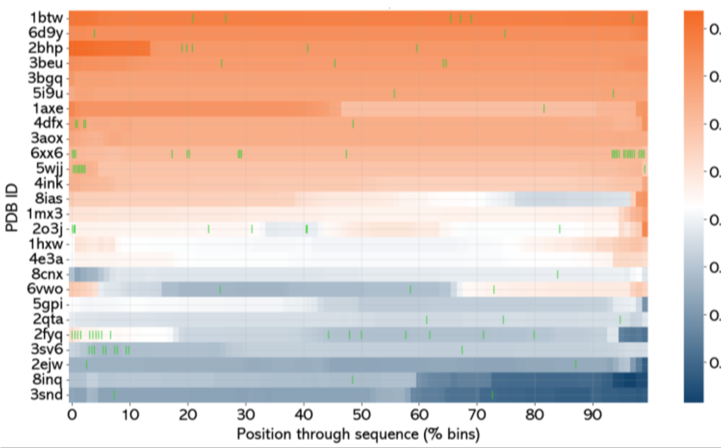


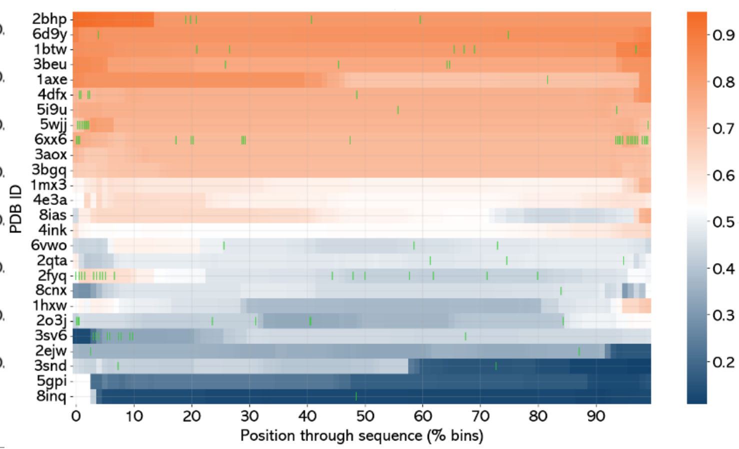
Figure 4.19. Change in hits, normalised as change in hits per residue.

The below chart (Figure 4.20) demonstrates the TM-alignment score of Foldseek across sequences normalised for length. Sequences are binned into 1% segments, and the average alignment score for all hits that cover that bin. Green lines represent the presence of homorepeats of at least 3 amino acids long. The third chart demonstrates the change in median TM-score against each segment of the sequence. Regions of declines in alignment quality are observable over many regions in sequences.

Wild-Type



Variant



Difference

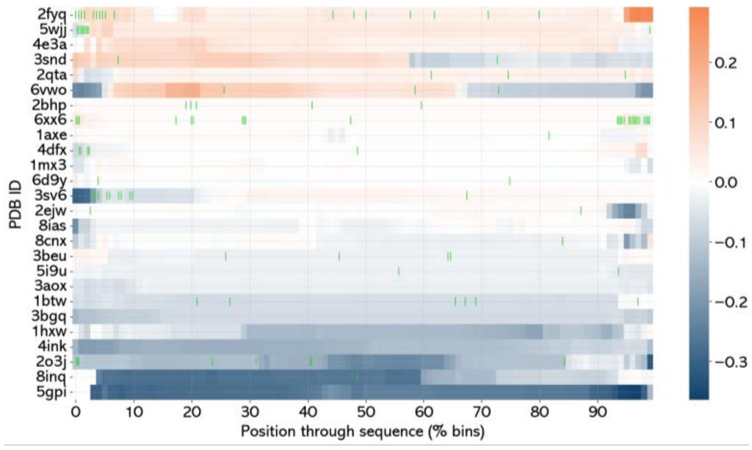


Figure 4.20. wild-type, variant, and difference heatmap of change in alignment quality (TM-score).

5. Discussion

Thermostability is a key feature that ESM2 optimises for

The core finding in these results of the analysis was that when a directed evolution process is applied to ESM2, and allowed to overrun until convergence, thermostability is a core feature for which the model optimises for. An overall change in melt temperature was observed in almost all proteins. In fact, this is something noted by Baker (2019): that protein stability is governed by the loss of configurable entropy, and the formation of attractive interactions, such as salt bridges and hydrophobic packing, and that higher stability is a common trait of *de novo* designed proteins. These traits are reflected in the amino acid substitutions chosen by the model.

Changes in amino acid propensity

Increases in glutamic acid and lysine in variants in this study support Baker's assessment of protein stability, in that they contribute towards salt bridges, a key contributor to the thermostability of proteins. However, salt bridge formation reduced in some proteins, and contacts reduced in most, despite thermostability and salt bridge forming residues increasing, suggesting degeneration of amino acid proposals. Additionally, a limitation of analysis by counting salt bridges, is that the contribution that a salt bridge has towards stability can vary by factors such as pH, distance between charges, charge geometry, and hydration shells, with the latter reducing the strength of solvated interactions, with buried salt bridges that are excluded from solvent being stronger as a result (Bosshard et al, 2004). By contrast, this pipeline used binary cut off by distance to count salt bridges without calculating their impact. Isoleucine and leucine were often substituted for valine, a smaller protein, facilitating denser hydrophobic packing, supporting optimisation for thermophilicity. Taylor & Vaisman (2010) performed a study that examined the mean propensities of amino acids in thermophiles, hyperthermophiles, and mesophiles. The propensity of amino acids from the variant set most closely matched thermophiles, followed by hyperthermophiles. Notably however, arginine propensity is significantly underrepresented in the variant group compared to thermophiles more broadly, which may be considered a key difference between genuine, and hallucinated thermophilic structures - this would, however, require testing with a much larger dataset across a wider variety of protein families to confirm. Additionally, there was no set pattern to which mutations occurred. Although terminal end mutations were common, this was not necessarily replicated consistently across the entire dataset, and in most cases, not particularly any more frequent than other regions. Only 5GPI, 3AOX, and 3SV6 experienced significant mutations localised entirely within certain regions.

Increases in propensity are often driven by homorepeat motifs

Given that salt bridge formation decreased in some cases, it may be the case that the model optimises for thermostability by increasing the proportion of residues that are associated with thermostability the creation of homorepeats, even if the context of these residues is due to positions and motifs which the model hasn't quite captured. This analogous within PLMs for a behaviour in text generation models described by Jin et al as degeneration – when LLMs can be stuck in repetitive loops. This was not just observed in salt bridge formation: it also occurred the way that the model tended to favour poly-E residues over motifs commonly overrepresented in thermophiles. Taylor & Vaisman (2010) noted that mesophilic and thermophilic organisms can be discriminated via key features in their sequences, in particular, the propensity of IVYWREL residues, and the ratio of EK/QH residues. The EK/QH ratio corroborated with this study: E and K residues were increased, whilst Q and H decreased, with only R having a major discrepancy in propensity for IVYWREL. An increase in EK would be expected in thermophilic proteins, due to the formation of salt bridges. Furthermore, it was also noted that there are a series of 4-member motifs that are heavily overrepresented in thermophiles. However, in this study, ESM2 had a lack of optimisation towards overrepresented motifs in thermophiles, rather, in many cases, particularly for glutamic acid, the model attempts to introduce repeats of the same amino acid, a degeneration hallucination. This analysis should be interpreted with caution however, as this may be an artefact of how EvoProtGrad allows for mutation rather than ESM2's predictions: the parameter for max mutations per MCMC step was set to 4, which may limit the possibility for EvoProtGrad to trial motifs, as opposed to more sporadically placed residues across the sequence. It is not just glutamate that frequently had homorepeats: notably, a frequent change was the removal of, and the addition of his-tags on termini. Long additions of homorepeats on terminal ends are not uncommon, demonstrated in the figures 4.18 and 4.20. The extent and distribution of these homorepeats throughout sequences can be observed in these figures too. This may suggest suggest that the model often enters degenerative proposal space in which it is proposing homorepeats, where substituting the same residue is an easy route to a higher log-likelihood for a sequence. In addition, when looking at figure 4.15., demonstrating common linker proteins used in biotechnology, very rarely is there an optimisation to a linker that may be overrepresented in databases, rather, homorepeats occur instead, with changes to same residue linkers occurring frequently. The exception to this is 3SV6 developing a GSGSG linker at the beginning of the sequence. To investigate this further however, it would be necessary to compute the predicted change in thermostability for MCMC steps that propose a sequence in which a homorepeat is extended.

ESM2 likely optimises for other features than thermostability.

Whilst increased thermostability is a dominant trait in the outputs of this study, it does not by any means suggest a model is hallucinated, Indeed, when a model confidently predicts a proposed sequence to be more likely, whilst we understand that increased thermostability is a key driver of this, we should also look for the features in the sequence that increase likelihood. This can result in some apparently contradictory observations. Salt bridges decreased in some proteins whilst thermostability increased, whilst contacts decreased, which would be at odds with increased thermostability. The most obvious explanation for this behaviour is that the model is optimising for other features. Indeed, often when optimising, a plateau in thermostability is observed, and in many cases, continues to decline, despite the model continuing to run for a large number of epochs, and is thus optimising for other features as a result. As ESM2 is trained on the breadth of evolutionary data, it is entirely possible that the model is exploring patterns learned by the model that are associated with increased log-likelihood in another region of protein space. Alternatively, it may be the case that increased thermostability is the dominant goal for the model, but it frequently makes incorrect decisions in when predicting for it – it is to be expected a model will make mistakes. The analysis of Hao et al (2025) showed that ESM2 650m could occasionally assign positive a LLR to a protein mutation experimentally to be deleterious, so it is feasible contradictory optimisations for thermostability can simply sit within the expected distribution of incorrect predictions. This may provide an explanation for apparently juxtaposed traits, such as increased thermostability but with reduced contacts, representing a hallucinated sequence that resulted in an implausible structure.

Folds in local alignment aligned poorly with folds found in nature.

Analysis of global and local alignments were used to see if there were any significant differences between the two methods in alignment to existing folds. Broadly, there wasn't a particularly significant difference the two methods. One of the simpler ways to identify whether a predicted structure has no biological activity would be to determine whether its folds are found in nature. The first sign that many folds were not found in nature was the significantly lower number of query hits normalised by sequence length in variant structures than wild-type structures in both global and local alignments. Secondly it can be observed that TM-score declined in many regions locally (figure 4.24), suggesting that these represent regions of sequences which did not align well to folds and structure found in nature. Despite the fact many sequences did in fact have broadly favourable TM-scores globally, pockets of reduced alignment when examined for local changes in TM-score elucidated these likely hallucinations. Furthermore, the green lines in figure 4.18 and 4.20

demonstrate homorepeat regions of low complexity and low specificity, further casting doubt on the quality of 3di strings that aligned to them. These areas of poor local alignment confirm the presence of implausible areas of structure within a protein structure that may otherwise possess a high level of structural similarity to functional and known proteins. This corroborates with the observation of Johnson et al (2024) that pLDDT and sequence similarity/identity to known sequence are poor predictor of biological function. Furthermore, as the models were generated with Alphafold2, which relies on MSA as a means of predicting structure, the fact these models had worse clashscores after being mutated for a long time corroborates with the reduced Foldseek hits – the MSA did not have as much depth when trying to align the variant sequences with known ones.

Common features in hallucinated proteins: overview

What does a "perfect" sequence look like for ESM2? Broadly, the sequences presented in this analysis, are all sequences that ESM2 could not, or could rarely improve anymore, and the collection of features in the below table represent common traits in the set of proteins mutated in this study. In conjunction with these though, the sequences predicted broadly resulted decrease in structural quality from sequences which had run until convergence. No hallucinated protein can be defined by a single common trait, however, there are collections of traits identified in this study associated with them. 'Quasi-instrinsic' is used to describe hallucinations here that are *likely* to be contradicting their source content - known biology of proteins:

Trait	Description
Increased thermostability	Not necessarily a predictor of hallucination, as significant increases in thermostability occur in early epochs during optimisation, but almost all hallucinated proteins continued to attempt to optimise for hallucination.
Reduction in alignment score in Foldseek	Reductions in TM-score on Foldseek queries are very common in proteins in the variant set, slightly more so on global alignments than local. This suggests that hallucinated protein structure becomes slightly less homologous to a template it is mutated from, whilst regions of local order remain in smaller folds.
Lower than average arginine propensity for thermophilic protein	Quasi-intrinsic hallucination. Whilst ESM2 optimises towards an amino acid propensity similar to that observed in thermophiles, with some traits of hyperthermophiles, there are a series of amino acids in which there is a larger difference in propensity, particularly arginine, which is underrepresented in the variant set.
Presence of homorepeat motifs, particularly poly-E	Degenerate hallucination. Whilst homorepeats occur in nature, they frequently get optimised for in ESM2. Changes to poly-H occur, likely due to overrepresentation of his-tags. Poly-E and poly-A motifs appear frequently in

	hallucinated sequences. ESM2 appears to optimise less towards particular motifs, but instead to attempt to increase propensity of amino acids that compose the motifs. The move towards poly-X repeats represents a collapse of complexity in the sequence.
Significantly less hits on Foldseek than wild-type	Structures generated from ESM2 appear to get less hits than wild-type comparators, this effect is more pronounced in certain proteins than others. In directed evolution processes, it may be helpful to pair the variant query with the structure it was mutated from. This discrepancy may be even more pronounced than laid out in this paper, due to quality filtration steps in Foldseek's algorithm.
Decreased contacts in tandem with increased thermostability	Quasi-intrinsic hallucination. Contacts being decreased in models predicted from variant sequences that have increased thermostability is an apparently contradictory feature, and may represent an implausible structure.

Further Research

Whilst the analysis in this paper cast out a wide net in search of features that were features of hallucinations, this could certainly be expanded on, particularly to search through unexplored protein space to catalogue other features. It would be worth performing comparisons between each proposed sequence in each epoch of the directed evolution loop, as the analysis laid out here only shows a wild-type vs variant comparison for amino acid substitutions. There is rather significant factor for this – whilst decisions that increase log-likelihood often also result in an increase in thermostability, many increases in log-likelihood do not, and it is important to characterise these, and capturing the decision making at each step, as opposed to a before and after of wild-type and variant is vital to understanding the decision making of the model. This provides a more granular view of the model's decision making as it provides an overview of the change in log-likelihood that each amino acid change would have in the sequence at any given point. In addition, as contacts declined in most proteins, it would be prudent to observe the change in flexibility of predicted models – an increase in thermostability combined with increased flexibility could represent an implausible combination of features, which may be a predictor for a hallucinated structure. However, to reliably predict flexibility for computationally generated models, a molecular dynamics method with a measurement of a feature such as root mean square fluctuation (RMSF) may be preferred, but this is computationally expensive and was not within the scope or scale of this project. Many analyses presented, such as location of mutations and contact changes did not find a significant pattern occurring in the placement of mutations and contact changes. However, this may be in part due to the limited sample size and selection of families. It may remain to be seen that defined patterns of mutations may manifest over larger data sets, and across a wider range of protein families if this study were to be scaled up. Additionally, all the protein families in this study were globular proteins, and it may stand that optimisations may differ in fibrous proteins. The max mutations parameter in EvoProtGrad could be altered to be a percentage of the length of the protein, to ensure that analysis of mutations as a proportion of a set, or of a protein are not biased by length. As a parameter of 4 maximum mutations per MCMC step was set, this may have limited the proposal for ESM2 to propose moves towards motifs that increase the log-likelihood of a sequence, as opposed to amino acid substitutions that increase the likelihood of a sequence, due to a smaller window for proposal space. In an expansion of the study, attempting to use ESMfold and rerunning structural metrics with this may also yield different results, as the hallucinations in predicted sequences impacted Alphafold2 predictions due to poor alignments in the MSA step.

Additional evaluation metrics could be used to improve the sensitivity of the methodology in this paper. Johnson et al (2024) found that inverse folding models (ESM-IF, ProteinMPNN, MIF-ST), and language model likelihoods were strong predictors, and expansion of further research could utilise these metrics. Although there are ways in which the methodology could be improved given more time and computational resource, the next steps in the direction of research regardless, would be identifying what drives the model to make the hallmarks of hallucinations identified in this paper. Most importantly, perhaps understanding why ESM2 fails to propose common motifs, and instead produce homorepeats would elucidate hallucination mechanisms significantly. Perhaps one of the most viable ways to approach this is to identify knowledge neurons most responsible for specific knowledge and decisions within a model (Sato & Takagi 2025).

A central method for identifying the knowledge neurons in a PLM would be using Nori et al's (2023) method, in that performing ablation studies to isolate subsets of knowledge neurons, using activation and gradient-based methods to visualise and capture the information flow through the neural network. their analysis found that value neurons were not particularly valuable for knowledge expression in ESM2, compared to key neurons. From here, the proposal would be to take an approach using a method of attribution, to identify key neurons amongst these associated with the key features of hallucination identified in this study as a starting point. Whilst we could potentially identify knowledge neurons containing "true positives" (Dai et al 2021), in that they are defined by their ability to perform accurately in a classification task, then perhaps the inverse is a possibility - selecting neurons that improve the log-likelihood of a model by proposing features that result in hallucinations: is neurons that confidently assess a sequence to be more likely despite it being nonsense which are the most problematic. It is from this, that attempts to build classification tools from well-labelled, understood, and annotated data on hallucinations can be made.

6. Conclusions

In summary, the aim of this study was to identify features of hallucinated protein sequences from ESM2, so these features could be interrogated in further research on the drivers and information flow of hallucinations in the model architecture. This was achieved by utilising a gradient-based MCMC directed evolution programme to iteratively mutate proteins until their improvements in log-likelihood converged. The resultant sequences were analysed for a series of features that they optimised for, with thermostability being the most dominant. Structures were generated from these sequences using Alphafold2, and it was found that these structures had a lower quality than models generated from variant sequences from early epochs. The results demonstrate a series of key features that occurred across the dataset that are likely to be present amongst hallucinations: increased thermostability, but with reduced contacts; homorepeat motifs, particularly glutamic acid, and particularly at the termini of sequences; lower than average arginine across a dataset of proteins; lower hits than a wild-type on Foldseek; areas of local poor alignment on Foldseek; a lack of common linkers thermophilic motifs, with the representation of amino acids from these motifs manifesting as homorepeats instead.

Whilst no single one of these features unequivocally substantiates evidence of a hallucinated structure, they crucially reduce the search space and set up further research on ablation studies and information flow within knowledge neurons to identify how ESM2, and even other models produce hallucinated sequences. This study sets the foundations for future work in improving the interpretability of PLMs, and the identification of hallucinations in generated protein sequences and structures. Furthermore, the results of this study open up avenues for the scrutiny of generated structures in molecular dynamics experiments, allowing for further comprehension of the effect hallucinations have on biological activity.

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Appendices

Code used and supplementary data available at: https://github.com/bareebaree/halluciTATE

Hydropathy, Aromaticity, instability index, aliphatic index, iso-electric point, net positive and net negative count were omitted from the final analysis for brevity. These are available in the attached results table however, and the notebook analysis script can be run on them. The counts for secondary structure proportion were based off a Biopython analysis off sequence. These were omitted due to use of Foldseek for finding folds, and DSSP being a superior method, which was omitted from the analysis for brevity. Flexibility on Biopython was originally captured, but not used in the analysis, due to the superiority of molecular dynamics methods, but their computational expense proscribed their use. What are presented in this paper are the most impactful results.